

Name:_____

Ans: 5 ★*START*

1

Add the polynomials:

$$\begin{aligned} &(4x^3 - 2x^2 + 5x - 1) \\ &+ (-x^3 + 6x^2 - 3x + 4) \\ &+ (2x^3 - x^2 - x - 7) \end{aligned}$$

Ans: 12

#_____

Simplify:

$$-2x^2(3x^2 - 4x + 1) + 5x^3$$

Ans: $3x^4 - 5x^3 + 7x^2 + 4x - 10$

#_____

Multiply:

$$(2x - 3)(x^2 + 4x - 5)$$

Ans: $x^2 + 5x - 4$

#_____

Multiply:

$$(x + 4)(x - 1)(x + 2)$$

Ans: $3x^3 - 2x + 1$

#_____

CHALLENGE: If $(x + 2)$ is a factor, what must the remainder be when dividing by $(x + 2)$?

Verify: $(x^3 + 4x^2 + x - 6) \div (x + 2)$

Quotient = ?

Ans: $3x^2 - 7x + 8$

#_____

Multiply (FOIL):

$$(3x - 4)(x + 2)$$

Ans: $3x^2 + 5x - 2$

#_____

$(2x^2 - 5x + 3) + (ax^2 + bx - 7)$ simplifies to $6x^2 - 2x - 4$.

Find a and b , then calculate $a \cdot b$.

Ans: $2x^2 - 3x + 1$

#_____

$(2x^2 - 3x + 1) - (x^2 + x - 4)$ simplifies to $ax^2 + bx + c$.

Find c .

Ans: $5x^3 + 3x^2 + x - 4$ #_____ Subtract: $(8x^4 - 3x^3 + 2x - 9)$ $- (5x^4 + 2x^3 - 7x^2 - 2x + 1)$	Ans: $-6x^4 + 13x^3 - 2x^2$ #_____ Divide: $(18x^5 - 12x^3 + 6x^2) \div 6x^2$
Ans: $2x^3 + 5x^2 - 22x + 15$ #_____ Divide: $(6x^3 + 7x^2 - 9x + 2) \div (2x - 1)$	Ans: $x^3 + 5x^2 + 2x - 8$ #_____ Divide: $(4x^3 - x^2 + 3x + 5) \div (x - 1)$ Write as: Quotient R Remainder
Ans: $x^2 + 2x - 3$ #_____ Subtract, then simplify: $\left[(3x^2 + 2x - 1) - (x^2 - 4x + 5) \right]$ $- (x^2 + x - 2)$	Ans: $3x^2 + 2x - 8$ #_____ Determine the difference of $7x^2 + 4x - 3$ and $2x^2 - x + 5$
Ans: $4x^2 + 3x + 6$ R 11 #_____ ERROR ANALYSIS: $(5x^2 - 3x + 2) - (2x^2 + 4x - 6)$ Maya's work: $5x^2 - 3x + 2 - 2x^2 + 4x - 6 = 3x^2 + x - 4$ Correct answer = ?	Ans: $5x^2 + 5x - 8$ #_____ Divide: $(10x^3 - 15x^2 + 5x) \div 5x$